

BOL'SHIYE UKI, Russian Federation

WMO: 284910

Lat: **56.933N**Lon: **72.667E**Elev: **87**StdP: **100.28**Time Zone: **6.00 (E06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.2	-32.2	-37.8	0.1	-35.1	-34.9	0.1	-32.1	6.5	-10.6	6.0	-11.2	0.9	170	0.608

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	28.6	20.1	27.1	19.0	25.5	18.3	21.1	27.1	20.1	25.6	19.0	24.2	2.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.9	24.3	17.9	13.0	23.1	16.9	12.2	21.9	61.6	27.3	58.0	25.6	54.5	24.2	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.9	6.0	5.3	DB	-38.8	31.7	3.9	1.7	-41.6	32.8	-43.9	33.8	-46.1	34.7	-48.9	35.9	(4)
(5)			WB	-38.9	22.7	3.9	1.5	-41.6	23.8	-43.9	24.7	-46.1	25.6	-48.9	26.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.2	-17.9	-14.9	-6.2	2.9	10.6	16.2	17.9	15.1	9.3	2.5	-7.5	-14.4	(6)	
	DBStd	13.80	8.05	7.44	5.82	5.67	5.45	4.59	3.68	3.85	4.39	4.96	7.70	8.17	(7)	
	HDD10.0	3938	866	697	502	221	62	7	0	4	66	236	524	756	(8)	
	HDD18.3	6336	1124	931	760	463	246	93	54	114	272	491	774	1014	(9)	
	CDD10.0	735	0	0	0	8	79	192	246	162	45	3	0	0	(10)	
	CDD18.3	91	0	0	0	0	5	29	42	14	1	0	0	0	(11)	
	CDH23.3	877	0	0	0	2	95	284	357	125	14	0	0	0	(12)	
	CDH26.7	174	0	0	0	0	21	62	68	23	1	0	0	0	(13)	
(14) Wind	WSAvg	2.4	2.2	2.3	2.7	2.8	2.8	2.2	2.0	2.0	2.2	2.5	2.6	2.5	(14)	
(15) Precipitation	PrecAvg	493	22	17	22	27	41	68	73	69	48	38	36	32	(15)	
	PrecMax	665	45	40	50	63	108	145	154	110	101	87	63	60	(16)	
	PrecMin	331	2	1	0	2	11	22	29	28	8	6	8	13	(17)	
	PrecStd	73	12	8	14	16	22	32	31	24	24	19	14	13	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.0	2.0	7.9	21.5	29.3	30.6	30.5	29.2	25.2	17.1	6.4	1.8	(19)	
		MCWB	-0.9	0.1	3.1	13.1	18.1	20.3	22.1	21.1	16.4	10.5	4.5	1.0	(20)	
	2%	DB	-1.9	-0.2	4.8	17.5	25.8	28.6	28.5	26.5	21.7	14.5	4.2	0.3	(21)	
		MCWB	-2.6	-1.4	1.7	10.4	16.3	19.8	20.6	19.2	14.6	9.3	3.0	-0.6	(22)	
	5%	DB	-4.0	-2.2	2.9	14.5	23.4	26.7	27.0	24.4	18.9	11.9	2.8	-1.6	(23)	
		MCWB	-4.7	-3.2	0.5	8.6	14.8	18.8	19.4	18.1	13.3	8.0	1.7	-2.4	(24)	
	10%	DB	-6.3	-4.4	1.5	11.9	20.6	24.7	25.4	22.3	16.7	9.7	1.3	-3.6	(25)	
		MCWB	-6.9	-5.4	-0.3	7.1	13.2	17.9	19.0	17.1	12.0	6.8	0.4	-4.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.8	0.4	3.6	13.2	18.9	22.2	22.7	21.6	17.4	11.7	4.9	1.2	(27)	
		MCDB	-0.3	1.8	7.0	20.9	27.6	28.3	28.8	27.4	23.2	15.2	5.9	1.8	(28)	
	2%	WB	-2.6	-1.4	2.0	10.9	16.7	20.6	21.4	20.0	15.3	9.9	3.2	-0.6	(29)	
		MCDB	-2.0	-0.3	4.2	16.7	24.8	26.7	27.4	25.1	20.3	13.5	4.1	0.1	(30)	
	5%	WB	-4.6	-3.2	0.9	9.0	15.3	19.5	20.4	18.7	13.9	8.4	1.8	-2.3	(31)	
		MCDB	-4.0	-2.2	2.6	13.9	22.3	25.6	25.8	23.1	18.2	11.3	2.7	-1.6	(32)	
	10%	WB	-6.8	-5.4	-0.2	7.3	13.7	18.4	19.4	17.6	12.4	7.0	0.5	-4.3	(33)	
		MCDB	-6.3	-4.5	1.5	11.5	19.6	23.8	24.3	21.6	16.1	9.7	1.2	-3.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	9.4	9.3	9.4	11.4	10.4	9.8	9.4	8.8	6.7	6.1	7.2	(35)	
		MCDBR	6.9	7.6	8.9	13.4	15.6	13.6	12.4	13.1	12.9	10.9	5.3	6.4	(36)	
	5% WB	MCWBR	6.7	6.9	6.6	8.1	8.0	6.7	5.8	6.6	7.2	7.0	4.8	6.2	(37)	
		MCWBR	6.9	7.4	7.7	12.7	14.4	12.4	11.3	11.3	9.9	5.1	6.5	6.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.275	0.311	0.368	0.413	0.405	0.408	0.426	0.424	0.372	0.351	0.309	0.279	(40)	
		taud	2.030	2.015	1.928	2.039	2.228	2.287	2.235	2.214	2.322	2.220	2.077	1.971	(41)	
	Ebn at Noon		566	706	754	790	830	833	805	777	769	663	512	431	(42)	
		Edh at Noon	61	96	137	144	127	122	126	120	93	79	59	49	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.75	1.77	3.26	4.50	5.22	5.67	5.30	4.13	2.72	1.36	0.62	0.45	(44)	
	RadStd		0.07	0.12	0.23	0.37	0.44	0.45	0.42	0.38	0.29	0.16	0.11	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page